DEEP Metrics

Northeastern Highlands reference assessment

Stream	(unnamed reach)
Coordinates	43.7, -72.2
COMID / HUC8	9327042 / 01080104
Drainage area	3.0 km2
Reach length	1234.5 ft
Watershed basis	NHDPlus V2 basin of the nearest StreamCat reach (StreamCat lookup engine)
StreamCat reach	COMID 9327042 (this reach)

Function	Metric	F/D	How it is measured	Status	Value	Source
Catchment hydrology	Impervious surface (%)	D	Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement.	Available	3.0	NLCD 2021
Catchment hydrology	Agriculture, crop and hay (%)	D	Percent of the delineated watershed in cultivated crops plus hay and pasture, computed by DEEP. No field measurement.	Available	12.75	EPA StreamCat
Surface water storage	Wetland (%)	D	Percent of the watershed in woody plus herbaceous wetland (NLCD 2019), read from EPA StreamCat for the reach. No field measurement.	Reference only	0.4	Computed by DEEP
Reach inflow	Road density (km/sq km)	D	Road length per square kilometer of the delineated watershed, computed by DEEP. No field measurement.	Available	3.0	NLCD 2021
Streamflow regime	Base flow index (%)	D	Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement.	Available	12.75	EPA StreamCat
Streamflow regime	Degree of regulation (%)	D	Normal storage of upstream dams as a percent of mean annual runoff, both read from EPA StreamCat for the reach. No field measurement.	Available	0.4	Computed by DEEP
Low flow and baseflow dynamics	Base flow index (%)	D	Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement.	Available	12.75	EPA StreamCat
Low flow and baseflow dynamics	Bankfull width:depth ratio	F	At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).	Entered	7.5	
High flow dynamics	Impervious surface (%)	D	Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement.	Available	3.0	NLCD 2021
Floodplain connectivity	Bankfull width:depth ratio	F	At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).	Entered	7.5	
Floodplain connectivity	Bankfull height above channel (m)	F	At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights.	*Measure in the field*		

Function	Metric	F/D	How it is measured	Status	Value	Source
Hyporheic connectivity	Sand + fines (%)	F	Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer.	*Measure in the field*		
Hyporheic connectivity	Embeddedness (%)	F	At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates.	*Not applicable*		
Channel evolution	Bankfull width:depth ratio	F	At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).	Entered	7.5	
Channel evolution	Bankfull height above channel (m)	F	At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights.	*Measure in the field*		
Channel and floodplain dynamics	Sinuosity	F	Take a compass backsight bearing and the distance between each pair of neighboring transects. Value: channel length along those segments divided by the straight-line distance between the reach ends.	*Measure in the field*		
Channel and floodplain dynamics	Bank angle (degrees)	F	On both banks of each transect, lay a pole about 1 m long against the bank and read its angle from horizontal with a clinometer, in degrees. Value: mean of the 22 readings.	Entered	12.5	
Sediment continuity	Relative bed stability (log)	F	Computed: the base 10 log of the geometric mean particle diameter divided by the largest diameter the bankfull flow can move, from the particle count, reach slope, bankfull depth, wood tally and residual pools (NRSA physical habitat calculations).	*Measure in the field*		
Sediment continuity	Sand + fines (%)	F	Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer.	*Measure in the field*		
Bed composition and bedform dynamics	Substrate size (log mm)	F	From the 105-particle size class count (as for sand and fines). Value: base 10 log of the geometric mean particle diameter in mm, from the size classes (NRSA physical habitat calculations).	*Measure in the field*		
Bed composition and bedform dynamics	Large wood volume (m3/100m)	F	Between each pair of transects, tally every piece of wood at least 10 cm in diameter and 1.5 m long in or above the bankfull channel, by length and diameter class. Value: wood volume per 100 m of channel, from the tally.	*Measure in the field*		
Light & thermal regime	Chlorophyll a (ug/L)	F	Water-column chlorophyll a (ug/L) from a 2 L mid-channel grab sample at the X-site, collected in an amber bottle, filtered and analyzed in a lab.	*Measure in the field*		
Light & thermal regime	Canopy density, mid-channel (%)	F	At mid-channel on each transect, read a convex spherical densiometer facing upstream, downstream, left and right (covered points of 17). Value: mean of the 44 readings, as a percent.	Entered	17.5	
Carbon processing	Riparian veg cover, woody (frac)	F	In a 10 m by 10 m plot on each bank at each transect, estimate the areal cover class of woody vegetation in the canopy, understory and ground layer. Value: summed woody cover of the 3 layers, averaged over the 22 plots, as a fraction.	*Measure in the field*		

Function	Metric	F/D	How it is measured	Status	Value	Source
Nutrient cycling	Total nitrogen (mg N/L)	F	Total nitrogen (mg N/L), lab analysis of the unfiltered mid-channel grab sample from the X-site, kept on ice.	*Measure in the field*		
Nutrient cycling	Total phosphorus (ug/L)	F	Total phosphorus (ug/L), lab analysis of the mid-channel grab sample from the X-site, kept on ice.	*Measure in the field*		
Water & soil quality	Specific conductivity (uS/cm)	F	Specific conductance (uS/cm). NRSA reports the lab value of a mid-channel grab sample at the X-site (the reach center); a calibrated multi-parameter meter reads the same quantity in the stream there.	*Measure in the field*		
Water & soil quality	Turbidity (NTU)	F	Turbidity (NTU), lab analysis of the mid-channel grab sample from the X-site. Sample before anyone walks the channel upstream.	Entered	22.5	
Habitat provision	Fast-water / riffle habitat (%)	F	At each thalweg station, record the channel unit (falls, cascade, rapid, riffle, glide, pool, dry). Value: percent of stations in falls, cascade, rapid or riffle.	*Measure in the field*		
Habitat provision	Natural fish cover (frac)	F	In the 10 m of channel centered on each transect, estimate the areal cover class of each fish cover type on the form (wood, brush, overhanging vegetation, undercut banks, boulders, others). Value: summed cover of the natural types, averaged over the transects.	*Not applicable*		
Population support	Benthic total taxa richness	F	From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: total number of taxa.	*Measure in the field*		
Community dynamics	Shannon diversity (H') (H')	F	From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: Shannon diversity of the taxa counted.	*Measure in the field*		
Community dynamics	Tolerant individuals (%) ~ HBI	F	From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: percent of individuals in pollution-tolerant taxa.	Entered	27.5	
Watershed connectivity	Mapped dams within one mile (dams)	D	Dams in the USACE National Inventory of Dams within 1 mile of the site, counted by DEEP. Note in the field any barrier the inventory does not show.	Available	3.0	NLCD 2021
Population support	Native fish taxa richness			*Not scored*		Insufficient reference support
Low flow and baseflow dynamics	Residual pool depth (cm)			*Not scored*		Insufficient reference support

F is measured in the field and D is answered from the desk. Available: the value was pulled and is printed on the field worksheet. Unavailable: no source answered, so measure or estimate it on site. Reference only: the value is shown and kept out of the score, because the curve was fitted on another data source. Not scored: the assessment withholds the metric for insufficient reference support.